

Stat 322/522: Study Design

Project 1

For this project assignment, you design, collect, and analyze data from a survey that you run. You work in pairs on the project, so that you can split the effort from data collection. Each member of a team is from the same course, e.g., students in STA 322 should work with other students in STA 322. If you cannot find someone else to work with, contact Prof. Reiter as soon as possible.

Your work should comprise a written report that describes the survey design and analysis. Please use R Markdown or some other word processor. Turn in one report per team, with all team members' names on the front page of the report. Include your code with the report. You can upload the report to one of your team members' Project 1 folder on our Canvas site. Additionally, please upload a csv file that includes the data that you used for the report. Reports that do not include the code or data will receive a lower grade. Please refer to the course website for the Duke Community Standard requirements for the project.

Due Date: Turn in your project to the Project 1 assignment folder on the Canvas course site by 7:00 PM on October 9. Please send me an email indicating which project partner turned in the report.

Grading: Maximum of 10 points. Grade will be based on appropriateness of survey design and analysis for the problem, clear justification of your design and analysis choices, and clear explanation of results.

Topic of the Survey: In this project, we use a survey to learn about the characteristics of Duke faculty in Trinity College. As a sampling frame, you will use departmental websites. You can look at scholars.duke.edu for information on individual faculty members. Please do not scrape the data for all faculty at Duke, as this defeats the purpose of practicing techniques in survey sampling.

Answer the following questions in your report, using point estimates and 95% confidence intervals.

1. What is an estimate of the fraction of faculty who are women? Design your survey with the goal of answering this question accurately. Note: use your own determination of each faculty member's sex based on the information in Faculty Scholars.
2. What fraction of faculty have at least one grant that is ongoing in 2026?
3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?
4. Answer one other question that interests you from the data that you could collect from scholars.duke.edu.

When specifying the design, consider ways to make the estimates as accurate as possible, e.g., stratified sampling or probability proportional to size sampling, and the data collection as convenient as possible, e.g., cluster sampling. You don't get extra points for being overly complicated in the design. Rather, use what we have learned to help you collect data with a time-efficient probability sample that allows you to get unbiased estimates. You should use a probability sample, even if you could figure out how to scrape all the data to get the actual population quantities. I leave sample size up to you depending on your resources (in this case, your time) and desire for accuracy, but I suggest sampling at least 75 faculty. With all project team members working together, you should be able to take a decent sample size without spending too much time on data collection.